

Virtual Mitosis Lab

Name _____

Find the virtual lab at:

http://www.biology.arizona.edu/cell_bio/activities/cell_cycle/cell_cycle.html

Answer the questions below as you read the first two pages of the virtual lab website.

1. Why are root tips so useful for observing mitosis? _____

Root tips contain many actively dividing meristem cells, so mitosis is easy to see.

2a. Describe what happens to the DNA in interphase.

DNA replicates in S phase of interphase, creating sister chromatids.

2b. Are the chromosomes visible? No.

3a. In what phase do the chromosomes become visible? Prophase.

b. What happens to the nuclear membrane at this point? The nuclear membrane breaks down.

4. During metaphase why are the chromosomes organized to line up in the middle of the cell?

Chromosomes line up in the middle so each daughter cell gets one copy of each chromosome.

This helps prevent chromosome-number errors during separation.

5. What is the main characteristic of anaphase? Sister chromatids separate and move toward opposite poles.

6a. What are two specific events that occur in telophase? Nuclear membranes reform and chromosomes decondense. Cytokinesis begins.

Two daughter nuclei form.

➤ Click NEXT at the bottom of the page. You do not need to copy the data table onto separate paper, as it's included here.

➤ Now you will begin the lab. You will be presented with 36 onion root tip cells. Determine which phase each cell is in. (You will know if you are correct or not.) Place a tally mark in the appropriate column in the table below. You should have a total of 36 tally marks when you are finished.

Interphase	Prophase	Metaphase	Anaphase	Telophase

➤ Complete the data table below by following the instructions for each row.

Data Table

	Interphase	Prophase	Metaphase	Anaphase	Telophase	Totals
# of cells	20	10	3	2	1	36
% of cells	55.56%	27.78%	8.33%	5.56%	2.78%	100%
# of minutes	588.9	294.4	88.3	58.9	29.4	1060

You'll need a calculator to complete some of the rows in this table.

- To fill in the **"# of cells"** row, count the number of cells in each of the phases from the data table on page 1 of this lab and write the number in the appropriate column.
- To fill in the **"% of cells"** row, divide the number of cells in each phase by 36 and then multiply by 100. (We use 36 because that's the total number of cells you observed.)
- To fill in the **"# of minutes"** row you'll determine how long a cell spends in each phase. To do this, multiply the % of cells by 1060, which is the total number of minutes in an onion root tip cell cycle.
- You can check your work for each row by adding up the numbers in each column. The total should be exact in the # of cells row, and extremely close, if not exact, in the other two rows.

Analysis Questions

1. Which phase of the cell cycle had the most cells? _____. Why do you think this is so? Interphase had the most cells because cells spend most of their time growing and copying DNA.

Interphase cells show diffuse chromatin and an intact nucleus.

2. Describe how the cells in interphase looked compared to the cells in prophase.

Prophase cells show condensed visible chromosomes and nuclear membrane breakdown.

3. Describe how the cells in metaphase looked compared to the cells in anaphase.

Metaphase: chromosomes line up at the center. Anaphase: sister chromatids separate.

Chromatids move toward opposite poles in anaphase.

4. Are the daughter cells produced by mitosis identical to their parent cell or are they different?

Explain your answer. Daughter cells are genetically identical to the parent cell (except rare mutations).

They also keep the same chromosome number as the parent cell.

5. Label the illustration below using the following terms:

Centromere: pinched middle where chromatids attach.

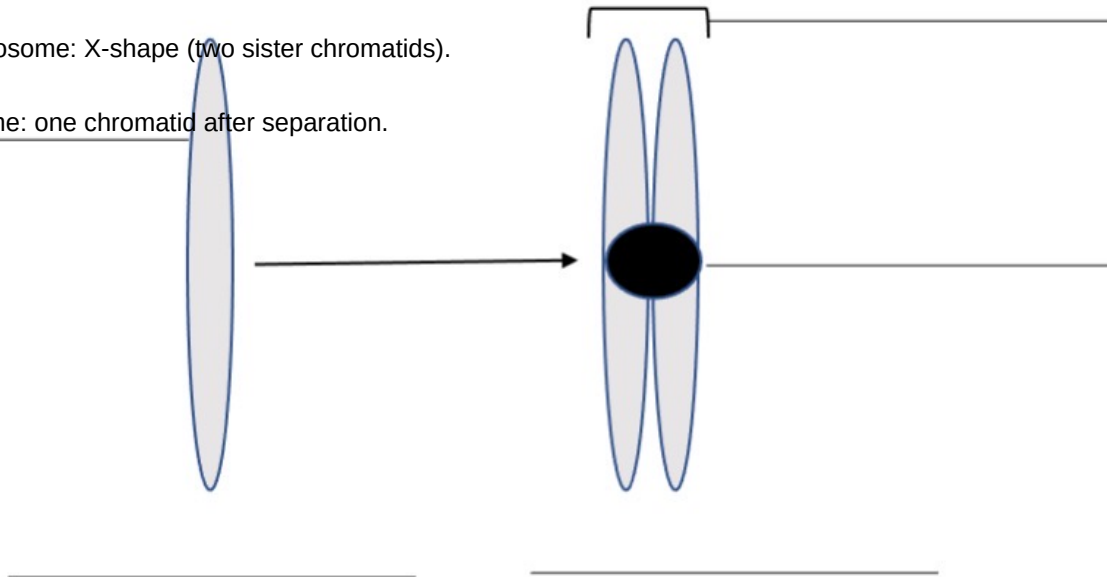
centromere, chromatid, sister chromatids, replicated chromosome, single chromosome

Chromatid: one strand of a replicated chromosome.

Sister chromatids: identical joined pair.

Replicated chromosome: X-shape (two sister chromatids).

Single chromosome: one chromatid after separation.



6. What process uses a single chromosome to copy itself?

DNA replication (S phase of interphase).

7. How many times do the chromosomes replicate in mitosis?

Once.

8. How many daughter cells are produced during mitosis?

Two daughter cells.

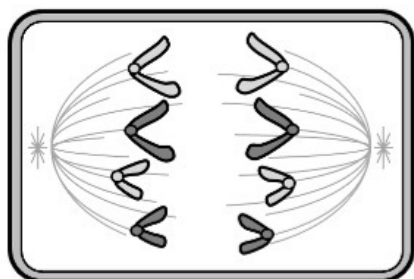
9. What is the term used to describe the "pinching in" of the cell membrane during telophase?

Cytokinesis (formation of cleavage furrow).

10. In what phase do the chromosomes replicate?

Interphase (specifically S phase).

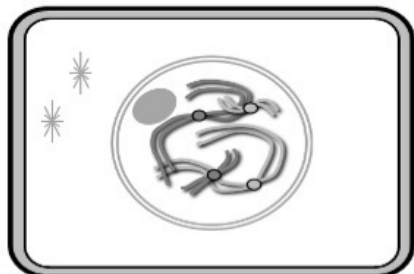
Label the phase each cell represents. Then describe the characteristics of each cell that helped you decide.



A.

A. Metaphase

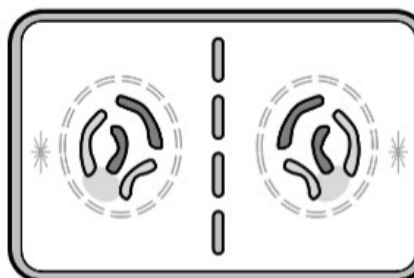
Chromosomes lined up across the center (metaphase plate).



B.

B. Interphase

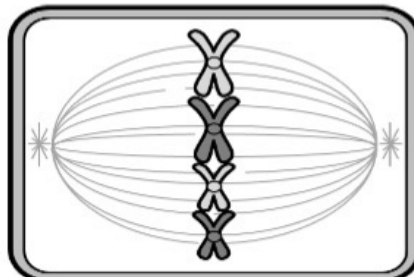
Intact nucleus and diffuse chromatin; no lined-up chromosomes.



C.

C. Telophase / Cytokinesis

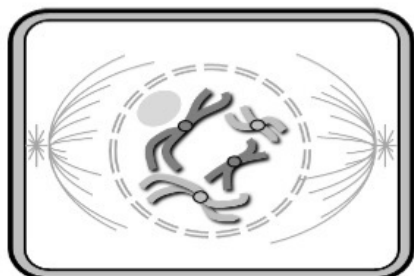
Two nuclei reform and the membrane pinches inward.



D.

D. Metaphase

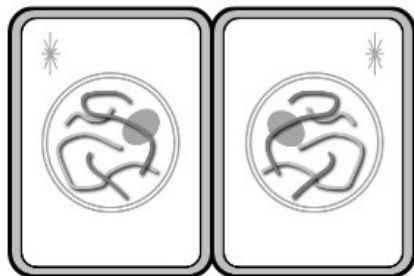
Chromosomes arranged at the middle of the cell.



E.

E. Prophase

Chromosomes condense; spindle forms; nucleus starts breaking down.



F.

F. Telophase

Chromosomes decondense and daughter nuclei are visible.